

Anchored Multi-Omics Integration and Knowledge-Anchored Residual Discovery

A never-below-the-best-view integrator, and a tool to find what the known biology misses — a white paper

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2026

A white paper for a technical-but-general audience. It distills the manuscript and its companion NotebookLM materials (briefing, study guide, data table, slide deck, infographic, and the transcribed audio overview and explainer video) into a single accessible document. All quantitative claims match the recorded result files and the manuscript.

Executive summary

Combining many kinds of molecular data is widely assumed to improve cancer prediction, yet in practice a single strong data type is hard to beat and naive fusion often does worse. This white paper presents an **anchored integrator**: it starts from established biological knowledge — a known high-performing data type, or a fixed, zero-parameter gene signature — and admits genome-wide data only as a non-negative gated residual, so the model is *provably never below its anchor* and improves on it only where another view carries orthogonal signal.

Three ideas follow from that one move. First, mining the **residual** (what the anchor cannot explain) turns the integrator into a discovery engine — a targeted search for the biology the known prior misses. Second, running it **in reverse** lets a hypothesis be tested against the prior and labelled novel, already-explained, or absent. Third, the framework makes **transportability** explicit: it shows precisely when, and why, the same biological signal looks novel in one cohort and redundant in another — often a property of the measurement technology rather than the biology. The result is a discovery engine that is honest about what is new and what the known biology already explains.

1. The problem: more data is not automatically better

The prevailing instinct in bioinformatics is to gather more — more scans, deeper sequencing, more molecular layers — on the assumption that fusing them yields clearer answers. The data say otherwise. A large benchmark across 14 cancers found that naively concatenating data types most often *hurt* survival prediction: stacking layers adds statistical variance that drowns the clear signal in a strong modality, and the model spends its capacity fitting noise. The honest objective is therefore not “always fuse,” but *never do worse than the best single view, and improve on it only where another view carries signal the leader cannot*.

A kitchen analogy from the companion audio captures it: a great broth does not come from tipping every leftover into the pot; you start from a strong foundation and add an ingredient only if it genuinely improves the flavour.

2. The method: anchor, gate, mine the residual

- **Anchor** on the empirically strongest modality, or on a fixed **zero-parameter** prior — a textbook gene-signature score that has had *no* training on the current dataset.
- **Gate**. Admit secondary data as $\text{logit}(\text{anchor}) + \beta \cdot \text{secondary}$, with the weight $\beta = 0$ chosen on held-out data. If the new layer is redundant or noisy, the gate stays shut ($\beta = 0$). This guarantees the model is never below its anchor and yields the framework’s golden rule: *never below the best single view*.
- **Mine the residual**. Rank features by partial correlation with the outcome while controlling for the anchor, surfacing **anchor-orthogonal** axes, with a matched-random-panel noise control and held-out / permutation verification.

On TCGA breast cancer, blind anchor selection routed correctly (methylation for methylation-defined clusters, RNA for PAM50 calls), never falling below the leader; a constructed positive control (signal invisible to RNA) engaged the gate for +0.047 AUROC. The companion explainer frames the gate as a veteran clinician’s established knowledge, with genome-wide data an eager newcomer allowed to speak only when it adds something genuinely new.

3. Discovery: a verified basal/keratinization axis

Luminal A vs B breast cancer is, by convention, a proliferation distinction. A **20-gene proliferation index (zero trained parameters)** reaches AUROC **0.919** — close to a fully trained 1,500-gene model (0.942) — and gating genome-wide data onto this fixed prior reaches **0.947**, the cleanest super-additive gain in the study. Strikingly, a textbook rule with no training nearly matches a 1,500-gene model; the framework is, in effect, teaching the model to read the textbook first so its compute hunts only for what is unknown.

Mining that prior’s residual surfaces a **basal / squamous-lineage axis** (KRT5/14/17/6B, TP63, DSG3/DSC3, SOX10, COL17A1, KLK5/7/8; keratinization, Reactome $p \approx 8 \times 10^{-11}$), verified by held-out replication (10/10), recurrence of the core panel (10/10), and collapse under permuted labels. It reproduces externally in METABRIC ($\Delta +0.036$; 20/30 gene overlap, hypergeometric $p \approx 7 \times 10^{-27}$), is confirmed in the independent SCAN-B cohort (Section 6), and recovers the same squamous/keratinization program in TCGA lung cancer (LUAD vs LUSC; 10/30 overlap, $p \approx 3 \times 10^{-16}$) — the same genes in a different organ.

4. Generalisation across anchors, diseases, and feature types

- **A second knowledge anchor (HER2)**. Anchoring on the ERBB2 amplicon (incomplete in TCGA, 0.752) discovers a verified neuroendocrine/secretory + immune axis ($\Delta +0.054$); it does not reproduce in METABRIC because the amplicon anchor is near-complete there (0.997), leaving no residual — an interpretable negative.
- **A specificity control (ER)**. Anchoring ER status on a complete ER/luminal signature (0.938) discovers nothing ($\Delta -0.001$): the method refuses to manufacture an axis when the prior already suffices — an honest negative, the opposite of a hallucination.
- **A different disease and feature type (NSCLC immunotherapy)**. Anchored on the textbook biomarker tumour mutational burden (TMB) and given mutation/clinical features, the residual independently recovers the field’s other established biomarkers — PD-L1 (orthogonal to TMB), and EGFR and STK11 mutation (resistance) — $\Delta +0.061$ vs +0.006 for random panels ($p = 0.038$).

5. Hypothesis-as-anchor: confirm, explain-away, or refute

Run in reverse, a candidate hypothesis is gated onto the textbook anchor’s residual for a three-way verdict — **SUPPORTED** (adds beyond the prior), **EXPLAINED_BY_TEXTBOOK** (predicts alone but redundant once the prior is controlled), or **REFUTED** — operationalising a

mathematical form of peer review. A gate-free **commonality decomposition** (after Tonidandel & LeBreton) then partitions a hypothesis’s explained variance into the part *unique* to it versus the part *shared* with the anchor, and a mediation split shows how much of its effect runs through the anchor. This yields a label — **NOVEL**, **REDUNDANT** (collinear), or **INERT** — that separates a signal that is *absent* from one that is merely *redundant*. A 50-set hallmark screen is self-validating: proliferation-type hallmarks add ≈ 0 beyond the proliferation anchor, while estrogen-response programs are the supported orthogonal hits.

6. Transportability: what fails to reproduce, and why

The framework’s diagnostic contribution is to explain non-reproduction rather than merely report it. The estrogen-response hypothesis is **NOVEL** in TCGA (unique $R^2 = 0.038$) but **REDUNDANT** in METABRIC (redundancy = 1.00, 96 % mediated through proliferation) — collinear, not absent. A controlled sweep (hold both marginal effects fixed, vary only the anchor-hypothesis correlation) reproduces this: the same ER effect is 100 % NOVEL at TCGA’s correlation (+0.19) yet collapses into a collinear/suppression valley at METABRIC’s (−0.17). A second endpoint, HER2, confirms specificity (INERT in TCGA, REDUNDANT in METABRIC).

Across four endpoints and four columns — TCGA RNA-seq; TCGA Agilent (the *same patients*, different platform); METABRIC (independent microarray); and SCAN-B (independent RNA-seq, ~3,400 tumours) — two endpoints transport and two do not:

Endpoint (anchor → hypothesis)	TCGA RNA-seq	TCGA Agilent	METABRIC	SCAN-B	Transports?
Basal vs rest (basal → immune)	NOVEL	NOVEL	NOVEL	NOVEL	☐
ER status (ER → proliferation)	NOVEL	NOVEL	NOVEL	NOVEL	☐
LumA vs LumB (proliferation → ER)	NOVEL	REDUNDANT	REDUNDANT	NOVEL	☐
HER2 (amplicon → ER)	INERT	INERT	REDUNDANT	NOVEL	☐

For Luminal A/B the label tracks **measurement technology**: corr(proliferation, ER) is positive on both RNA-seq cohorts (+0.19, +0.10 → NOVEL) and negative on both microarrays (−0.10, −0.17 → REDUNDANT), flipping even on the *same* TCGA patients between their RNA-seq and Agilent measurements. As the companion media puts it, the same tumour viewed through “RNA-seq glasses” versus “microarray glasses” casts an opposite statistical shadow — so a model can end up *learning the platform, not the patient*. Genuinely anchor-orthogonal axes (basal→immune; ER-status→proliferation) are immune to this and transport across both platform and cohort.

7. Implications and conclusion

Predictive gains over a strong anchor are modest by design; the value is **routing and discovery** plus **diagnostic honesty**. The framework reports honest negatives (the basal axis marks

lineage identity, not survival, in the studied cohort), scales efficiently (485,577 methylation probes vs ER status in ~24 s, recovering the canonical *PGR* gene), and shows that external reproducibility tracks biological coherence: pathway-enriched, anchor-orthogonal axes reproduce across cohorts, platforms, and cancers, while collinear, measurement-dependent signals do not — and the framework now says which is which.

The broader caution lands beyond any single result: if a calibrated model’s verdict can flip with the assay, some apparent replication failures across the literature may be platform artefacts rather than biological disagreements. An anchored, residual-aware frame gives a rigorous way to tell the two apart.

Core philosophy: anchor on established biology and let a gate decide, honestly, whether genome-wide data beats it — and when a hypothesis fails to add, say whether it is novel-elsewhere, merely redundant, or truly absent.

Appendix A — Results at a glance

Endpoint / target	Anchor	Anchor AUROC	Residual gain (Δ AUROC)	Discovered axis	External status	Label
Luminal A vs B	20-gene proliferation	0.919	+0.029	basal/keratinized (KRT5/14/17/01; TP63, DSG3/DSC3, 10/30; SOX10, COL17A1, KLK5/7/8)	METABRIC (anchor near-complete 0.036; lung SCAN-B confirmed)	NOVEL
HER2 status	ERBB2 amplicon	0.752	+0.054	neuroendocrine + immune	not secretory — METABRIC (anchor near-complete 0.997)	—
ER status	ER/luminal signature	0.938	−0.001	none (specificity control)	n/a	INERT
NSCLC anti-PD-1 benefit	TMB	0.60	+0.061	PD-L1; EGFR/STK11 mutation	recovers established biomarkers	—

Appendix B — Companion media (NotebookLM Studio, transcribed)

Both generated from the same manuscript source and, on transcription, faithful to it (automatic-speech slips corrected: PD-L1, STK11, LeBreton, gene names). Full transcripts accompany this paper.

- **Audio overview** (~24 min, two-host deep dive): data-overload imagery; the broth analogy; the gate as a “taste test at the door”; the basal discovery and its METABRIC/lung/NSCLC validation; the ER specificity control as an honest negative; and the transportability trap (the proliferation-ER correlation flipping sign on the *same patients* across RNA-seq vs microarray, framed as two coloured glasses).

- **Explainer video** (~7.6 min), five parts — the multi-omics myth, the knowledge anchor, mining the residual, cross-cancer discoveries, and the platform trap; memorable framings include “never below the best single view,” a “search engine for the dark matter of the tumour’s biology,” and “learning the platform, not the patient.”

Appendix C — Data and code

All data are public and de-identified: TCGA (BRCA RNA-seq and Agilent, LUAD, LUSC, HNSC) via UCSC Xena; METABRIC and the NSCLC anti-PD-1 cohort via cBioPortal; SCAN-B via GEO accession GSE96058. Code, reproduce runners, recorded metrics, unit tests and CI guards live in the `omniomics` repository; the formal manuscript and preprint accompany this white paper.